Project Demo Planning

Date:	21 October 2026	**Team ID:**	Piyush-Balodiya-GenAI-2026
Project Name:	Multimodal Construction Intelligence Platform (CONSTRUCTAI)	**Maximum Marks:**	1 Mark

Project Demo Planning:

A well-structured demo plan ensures that the team presents the project effectively, covering all key aspects in a clear and organized manner. This document outlines the plan for demonstrating the project, including the flow of the demo, key features to highlight, and responsibilities of each team member.

Demo Section & Agenda Breakdown:

S. No	Demo Section	Description	Duration (mins)	Responsible Member
1	**Introduction & Problem Framing**	Overview of high fatal site accidents, manual inspection fatigue, and delayed structural crack repairs in civil projects.	2 mins	**Piyush Balodiya**
2	**Architecture & Tech Stack**	Explanation of FastAPI backend, React Vite high-contrast UI, and Groq LPU multimodal vision inference pipeline.	2 mins	**Piyush Balodiya**
3	**Live Demo: PPE Hazard Detection**	Live upload of CCTV snapshots showing sub-second detection of missing helmets, vests, and trigger of danger sirens.	3 mins	**Piyush Balodiya**
4	**Live Demo: IS 456 Crack Diagnostics**	Capturing surface wall cracks, classifying width severity, and generating automated epoxy repair prescriptions.	3 mins	**Piyush Balodiya**
5	**Live Demo: Weather, BOQ & PDF Export**	Demonstration of real-time rain risk advisory, automated material takeoffs, and instant 1-click audit PDF generation.	3 mins	**Piyush Balodiya**
6	**Q&A; Session & Technical Defense**	Addressing evaluator questions on latency benchmarks, zero cloud image logging, and IS 456 code integration.	2 mins	**Piyush Balodiya**

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Demo Flow Summary:

Structured sequential walkthrough planned for the evaluators, ensuring comprehensive coverage of problem definition, multimodal live demonstrations, and architectural defense.

Step	Activity	Detailed Notes & Execution Strategy
Step 1	**Introduction & Problem Statement**	• Highlight real-world industry pain points: multi-acre blind spots and delayed crack diagnostics. • Emphasize financial impact: regulatory OSHA penalties and concrete foundation washouts. • State project objective: Sub-second AI auditing without expensive recurring sensor hardware.
Step 2	**Solution Overview**	• Present CONSTRUCTAI architecture: FastAPI backend + Groq Cloud LLaMA 3.2 Vision LPUs. • Explain zero data retention security model ensuring private site images are never stored remotely. • Show high-contrast outdoor dashboard design optimized for field engineering tablets.
Step 3	**Live Feature Demonstration**	• **PPE Safety Run:** Drag & drop worker snapshot; observe sub-900ms bounding box detection and alarm playback. • **Structural Crack Run:** Upload beam crack photo; verify automatic IS 456 classification and epoxy repair guide. • **BOQ & Weather Run:** Query live rain radar thresholds and calculate cement/rebar quantity takeoffs. • **PDF Export:** Click export to instantly generate official executive audit summary PDF.
Step 4	**Q&A; Session**	• Defend architectural modularity, offline SQLite caching, and token throughput. • Discuss real-world scalability, drone camera integration, and enterprise ERP procurement compatibility. • Conclude demo with project roadmap and potential field test metrics.